

VR immersiveness and prevent VR-motion discomfort.

Rigorous compatibility testing aids in measuring performance of the app when accessed by different devices and platforms. It also helps identify dangerous non-functional issues or bugs, such as device overheating.

Usability is a quality attribute that indicates ease of use of a certain application. VR is an artificial environment created by the software. This way of perceiving the environment is completely immersive and supported by two essential physical devices, headsets and computer/console/cellphone, which create the virtual environment.

Pre-production testing checks whether the use-case is aligned with preferences and needs of the target audience.

A good quality VR experience requires high consistent frame rate, low latency between head motion and display output, and frames that have to be pre-distorted to offset the effect of looking at the display through an optical lens. To ensure a good VR experience, the tool captures and tracks the following key performance metrics:

Frame time. This is the time it takes to render each frame. The lower the frame time, the greater the possibility that the app will maintain a frame rate of 90fps.

Dropped frames. A frame drop occurs whenever the frame rendered by the VR game arrives too late for the headset to display. This causes the game to stutter and increases perceived latency, which can result in discomfort.

Warp misses. This occurs whenever a new or re-projected frame is not created in the current refresh interval. The user experiences this miss as a significant stutter.

Synthesised frames. If FCAT VR detects a lot of asynchronous space-warp frames, the system struggles to keep up with demands of the game. A synthesised frame is better than a dropped frame but is not as good as a rendered one.

Tools for VR testing

There are many tools available in the marketplace to test readiness of the PC for VR, as follows:

OptoFidelity BUDDY. It is a comprehensive solution for VR, AR,

mixed reality (MR) and head-mounted display (HMD) performance testing and calibration in both R&D and manufacturing lines. It comes with an integrated vision module and the desired number of degrees of freedom, which ensures the best HMD performance. It also measures audio-to-video synchronisation and motion-to-audio-latency for spatial audio.

FCAT VR. This helps measure performance of VR PC games and prevents stuttery unresponsive gameplay, which can lead to eye strain and discomfort. An FCAT VR analyser analyses performance and quality of the experience in a graphical user interface.

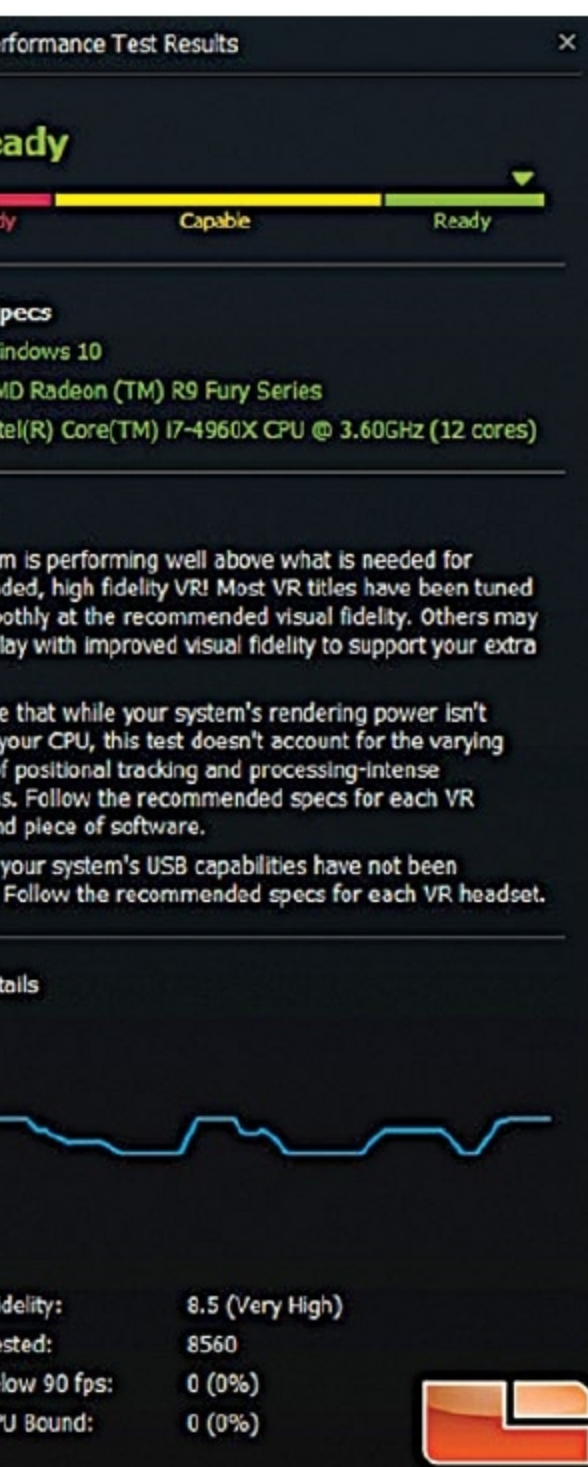
The common evaluation method is to experience VR and see how it feels in motion. As stutter and latency can degrade the experience to the point of motion sickness, these are critical to be measured while VR testing. FCAT VR provides comprehensive performance measurement for frame time and stutters on the VR headset without the need for special external capture hardware.

360° eyeTracker software solution. BluFocus has developed a unique VR positional tracking tool that allows analysing a user's behaviour within a 360-degree environment. It tracks position and orientation of the user's head movement while he or she is in a VR experience to extrapolate the field-of-view (FOV). This allows the user to see if visual, haptic and audible cues are directing his or her attention as desired by the storyteller.

Tools for VR app testing

The primary step in diagnosing performance problems is determining whether the app is CPU- or GPU-bound. VR app testing depends on specialised hardware. To make sure that VR hardware and app function properly, these must be tested on different devices and against all requirements.

VR headset manufacturers have



SteamVR performance test (Credit: www.legitreviews.com)

created a series of free tools to help potential customers make an informed choice. The tools measure frame time on a device and identify low-performance spikes. Some such tools are Oculus Rift Compatibility Tool, HTC VIVE Test Tool, SteamVR Performance Test and Microsoft Mixed Reality PC Check. One can simply download the tool and check the computer and display the result, indicating each component needed to play VR games.

Some other tools available in the market for VR testing are:

- Unity Profiler and Unreal Profiler for their respective engines
- Performance HUD for real-time monitoring of performance on device
- GAPID for monitoring specific graphics calls
- Android Studio's Systrace for detailed performance analysis **EFY**